

Uncovering the Olympic Games

Summary

The Olympic Games are a globally renowned sports event. This kind of attention has even driven the development of sports betting targeting the Olympic medal table, and the prediction of the medal table for the next Olympic Games has become a widely concerned research topic.

This paper analyzes the factors that may affect the number of medals a country wins, establishes a **discipline advantage level index system**, and combines a **two-step random forest model** - first performing binary classification on all countries first performing binary classification, and then conducting regression on the medal-winning countries. Then we use R^2 Score and RMSE to test the two-step random forest model. According to our prediction, the **United States should be the first country** on the Olympic medal table at the 2028 Los Angeles Olympics, **with 42 gold medals**, continuing its long-standing advantage. For the second sub-question, we compare the list of medal-winning countries obtained in the first step with the historical medal tables and find that **Czechia** is likely to win medals for the first time in 2029. Finally, by combining the Discipline Advantage Level Index system, the relationship between the host country's setting of sports events and the Discipline Advantage Level Index was explored, and a possible explanation for the host effect was discovered.

For the second question, combined with the **RDD (Regression Discontinuity Design)** in econometric models, the derived **RDiT (Regression Discontinuity Design)** model effectively proves the "great coach effect" and its impact on the number of medals. On this basis, we used the **K-means clustering algorithm** to help Australia, Italy, and Germany identify disciplines that are conducive to their potential development.

In the analysis, we also found that the athletic careers of most developing countries are shorter than those of developed countries, and the distribution of medals among countries and continents is significantly uneven. Athletes in developing regions have a faster iteration speed and need further support from the International Olympic Committee.

Keywords: discipline advantage level index system, two-step random forest model, RDiT, K-means clustering algorithm, Olympics

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1 Introduction

1.1 Problem Background

The Olympic Games represent a premier global sporting event that unites the world's most elite athletes, who compete not only for personal excellence but also for their nations' honor. Held every four years, this event draws top competitors from a wide range of sports disciplines, often serving as a platform to demonstrate the national strength of participating countries. Therefore, the rankings on the Olympic medal table have evolved into a measure of national power, becoming a highly valued metric for citizens globally. In this context, forecasting Olympic medal standings has gained significant appeal, with various organizations utilizing distinct models to make predictions. Accurate projections of the Olympic medal table play a critical role in shaping betting odds for lottery companies and guiding nations in strategically targeting their Olympic performance. As a result, the analysis and prediction of Olympic medal standings have garnered substantial attention in both academic and professional spheres.



Figure 1: Olympics medal table in 2020

1.2 Restatement of the Problems

Considering the background information and restricted conditions identified in the problem statement, we need to solve the following problems, utilizing the provided data on annual participation of athletes, host countries of the Olympics, medal distribution among nations, and variations in event settings across different years.

- Constructing a predictive model to forecasts the final medal standings for each country at the 2028 Los Angeles Olympics and evaluates the accuracy of these predictions.
- Examine the data to identify the "great coach" phenomenon and assess its impact on the medal standings. Select three countries and determine the sports in which they should consider investing in a "great" coach, estimating the magnitude of this effect.

- List some additional novel insights regarding Olympic medal counts from what your model had uncovered.

1.3 Literature Review

Predicting the Olympic medal table has long been a topic of great interest. Despite significant advancements in algorithmic methods, such as machine learning, these approaches often fall short in accurately forecasting Olympic medal table. In most cases, they even perform worse compared to the simple assumption that each country will win the same number of medals as they are in the previous Olympic Games.

- The earliest attempt to predict Olympic medal success was made by Ball (1972)[1], who introduced a correlation-based scoring model. Since then, forecasting models have evolved to become more sophisticated. Subsequent scholars employed ordinary least squares (OLS) regression to predict medal counts, but these methods failed to adequately address the issue of many countries winning zero medals. To better capture the distribution of Olympic medals, researchers later adopted Tobit regression models. However, even these advanced approaches are still underperformed relative to the naïve forecast.
- Building upon the insights from leading research in the field [2], the authors adopted a two-step modeling approach. This method first estimated the probability of a country winning any medals, and subsequently, in the event of success, predicted the exact number of medals. While this approach significantly improved the model's prediction accuracy, it still lagged behind the performance of the naïve forecast. To further enhance the predictive accuracy of the model, two strategies are considered: First, we conduct a comprehensive pre-screening of the training dataset to exclude any major political and economic events that could potentially distort the medal rankings. Second, we explore additional independent variables that could influence the likelihood of winning medals, as well as their potential impact on the final medal tally.
- Informed by the relevant literature[3], we have chosen to focus on data from 1992 onward. In selecting the independent variables for the model, we conducted an extensive review of existing research and carefully synthesized findings from numerous studies. Based on this review, we identified and selected five key explanatory variables: the size of the athlete delegation, the country's status as a host, geographical factors, medal performance in the previous Olympic Games, and the number of participants in sports traditionally associated with national strength. These variables were selected for their established relevance to Olympic outcomes in previous studies.

1.4 Overview of Our Work

2 Assumptions and Justifications

In reality , the factors that impact the result of the Olympics medal table can be diverse, and we need to make reasonable assumptions to simplify the model and clarify the input data's features ,and each hypothesis is closely followed by its corresponding explanation:

- Only data after 1992 can explain the Olympic medal table for 2028 .
- A country's strong sports disciplines remain unchanged after 1992 .
- The participating countries or regions in 2028 will be the same as in 2024, and the number of participants will also be equal.

3 Notations and Glossary

Symbol	Definition
G_{ij}	golds won by the country(remarked by j) in the i th Olympic Games
S_{ij}	silvers won by the country in the i_{th} Olympic Games
B_{ij}	bronzes won by the country in the i_{th} Olympic Games
P_{ij}	athlete participation in all events by the country in the i_{th} Olympic Games
E_{ij}	the number of all events in the i_{th} Olympic Games
$Host_{ij}$	Whether the country has the host effect in the i th Olympic Games
$SummaryIndex_j$	the country's capability of winning a medal

Table 1: Notations

4 Data Pre-processing

4.1 Data Chosen

Only data after 1992 is used in our model .To explain the reason behind it ,we insist that the number of Olympic medals is often a reflection of a nation's overall national power. But the period prior to the 1990's was characterized by dramatic shifts in the global order, including the two World Wars, the Cold War standoff, and the dissolution of the Soviet Union. These events had a profound impact on national power. Additionally, the number of participating countries varied greatly before 1992 due to several Olympic boycotts (such as the large-scale boycotts of the 1980 Moscow and 1984 Los Angeles Games, as well as the East German doping program, which accounted for 17% of the medals awarded to female athletes at the 1972 Games). These factors not only caused substantial fluctuations in the number of medals won by competing nations but also coincided with

geopolitical changes, such as the reorganization of national borders. These fluctuations and geopolitical shifts could significantly influence the predictive model. However, in the past three decades, the international environment has been relatively stable, with few disruptions like the decline of national power or the fragmentation of countries. Consequently, in order to align with current conditions and enhance the accuracy of predicting medal standings for the 2028 Olympics, we have opted to exclude data from the Olympic Games held before 1992.

We conducted preliminary data pre-processing for the data from 1992 on. Duplicate entries in the summer Olympic-athletes dataset were removed, and the names of countries were standardized and consolidated. Based on this dataset, we calculated the number of participants and medals won by each country annually, as well as their historical medal records across various sports disciplines. Additionally, we categorized the participating countries based on whether they were the host nation and their respective United Nations regional classification, transforming these criteria into categorical variables.

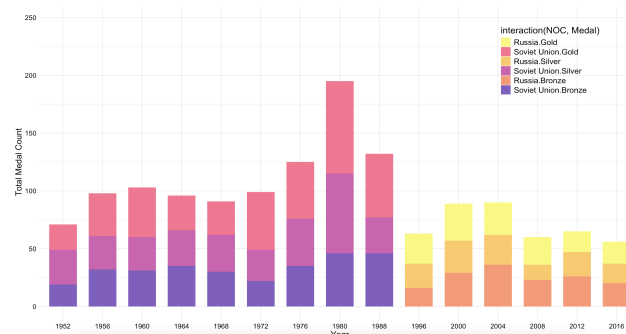


Figure 2: the Number of Total Medals Won by Union Soviet and Russia

4.2 Data Cleaning

We conducted preliminary data pre-processing for the period after 1992, removing duplicate entries in the `summerOly_athletes` dataset. Upon careful examination of the data provided in the tables, we found a substantial amount of information that required cleaning.

- Garbled code: Refer to the team abbreviations(NOC) to identify the countries.(e.g. Turkey, replace the original country name with a complex word) .We choose to solve this problem by replace the country's name with correct one .Hence, we need to consolidate and standardize the names of countries, and based on this table, tally the number of participants and medals won by each country in each year, as well as their medal-winning records in various events over the years.
- Different ways of remarking: the same country may have different names in different years . We could also use the NOC to unify the name of each country .

Name	Sex	Team	Noc	Medal
D'jamila Tavares	F	So Tom and Principe	STP	No Medal
Herminia Teixeira	F	Sao Tome & Principe	STP	No Medal

Table 2: dirty data example 1

Name	Sex	Team	Noc	Medal
Stephen Alford	M	United States	USA	Gold
Laurent Alfred	M	United States Virgin Islands	USA	No Medal
Brian Baker	M	United States-1	USA	No Medal
Thu Banh	F	United States-2	USA	No Medal
Gail (wolze-)	F	United States-3	USA	No Medal

Table 3: dirty data example 2

- Legacy issues from history (e.g. United States Virgin Islands, Great Britain Virgin Islands), make some places use additional names to remark its history. In these cases, we choose to determine by final belonging of the medal. For instance, Hong Kong's medal count is separated from the mainland, so we label it as a isolate region. But places like United States Virgin Islands are not the same, so we take it as a part of the US. in medal counts.

4.3 Data Processing

When predicting the performance of national teams at the Olympics, we need to first effectively pre-process the characteristic data of each country. The given dataset can be categorized into the following types of data based on features and content:

- The number of each kind of Olympic medals won by the country in the previous Olympic Games
- The total number of athlete participation in all events by the country in this year
- The total number of the events that the Olympic Games has
- Whether the country has held the Olympic Games before
- The country's sport ability estimated via discipline advantage level index

5 Problem Solution 1: Model 1

5.1 Discipline Advantage Level Index

To systematically and deterministically assess the disciplines with obvious advantages, general advantages, and potential advantages, we will conduct a stratified analysis based on the hierarchical structure of competitive sports.

Name	Sex	Team	Noc	Medal
Mesud Pezer	M	Bosnia & Herzegovina	BIH	No Medal
Emir Muratovic	M	Bosnia and Herzegovina	BIH	No Medal
Angelik Abebame	F	Congo (Brazzaville)	CGO	No Medal
Kongolo Amba	F	Congo (Kinshasa)	COD	No Medal
Stefan Sangala	M	Republic of the Congo	USA	No Medal
David Tshama	M	Democratic Republic of the Congo	COD	No Medal
Arnold Kisoka	M	DR Congo	COD	No Medal

Table 4: dirty data example 3

Name	Sex	Team	Noc	Medal
Angelik Abebame	F	Congo (Brazzaville)	CGO	No medal
Kongolo Amba	F	Congo (Kinshasa) Islands	COD	No Medal
Davor Tefanek	M	Serbia and Montenegro	SCG	No Medal
aba Silai	M	Serbia	SRB	No Medal
Goran Stojiljkovi	M	Montenegro	MNE	No Medal

Table 5: dirty data example 4

Different countries often have their own disciplines of expertise, and in a particular category of sports, they can often win many medals. In this case, in order to maximize the performance of the host country in the Olympic medal table, the host country is likely to increase the number of events in its obvious advantageous disciplines and reduce the number of events in potential advantageous disciplines of its main competitors, thereby effectively improving the ranking of the host country in the Olympic medal table. In this case, the organizer can influence the situation of Olympic medals by choosing the number of events. Therefore, we need to use certain methods to quantify and compare the advantageous disciplines between different countries. After referring to the literature on the advantageous events in which the Chinese team participated in the 2008 Olympics[4], we introduce , we introduce Discipline Advantage Level Index T_{jk} (k represents aa specific discipline). This study converts the rankings of a certain country in all Olympic Games over the past thirty years into scores according to the same formula below, and then calculates the strength level index of the country in this project. The calculation formula is:

$$Z_{jk} = 1 \sum_{i=1992}^{2024} G_{ijk} + 0.6 \sum_{i=1992}^{2024} S_{ijk} + 0.4 \sum_{i=1992}^{2024} B_{ijk} \quad (1)$$

$$T_{jk} = \frac{Z_{jk}}{\sum_{k=1992}^{2024} N_k} \quad (2)$$

Here, N_k denotes the total number of Olympic Games the country has participated in since 1992 . For each Olympic gold medal won, 1 points are awarded; for each silver medal, 0.6 points; and for each bronze medal, 0.4 points. That is how the total medal points are calculated according , **which also means that this index measures the average number of gold medals the country wins in this event per Olympic Games.**

We substitute the data of various countries into it to calculate the strength level index of different projects for each participating country. And identify the country's ability in this discipline by separating discipline advantage index as follows:

Standard	Definition
$T_{jk} < 1$	potential advantageous discipline
$1 \geq T_{jk} \leq 2$	general advantage discipline
$T_{jk} > 2$	obvious advantage discipline

Table 6: Caption

NOC	Discipline	Discipline Advantage Level Index
United States	Swimming	21.09
United States	Athletics	18.44
China	Diving	7.64
China	Weightlifting	7.10
China	Gymnastics	5.71
Australia	Swimming	12.56

Table 7: Example of Discipline Advantage Level Index

1.The relationship between event type and the number of medals a country wins:

Through the above analysis and research on the project level index, we can find that since the project index is calculated based on the number of medals won, it can be considered that the advantage level index of the country in this project calculated by us has practical significance. That is to say, the level index of this project can directly reflect the impact of this type of project on the total number of medals won by the country. Below are some images comparing the obvious advantages of major countries in different fields at the Olympics(The longest length indicates that the country has a relatively more obvious advantage in this sport) .

2.Explore the relationship between the number of projects and the number of medals obtained.

We have plotted a line chart for the number of events in each discipline at all the Olympics held since 1896 (as shown in the figure below) for preliminary data screening: exploring the relationship between the types of projects and years, among which relatively smooth projects are removed, and disciplines with obvious changes are retained .

Next, we will explore the relationship between the significant changes in the number of these four disciplines and the total medal counts of their corresponding dominant countries. (Swimming and Athletics - US, Shooting and Diving - CHN).

The results clearly show a positive correlation. This preliminarily indicates that there is a positive correlation between the increase in the number of the country's advantageous

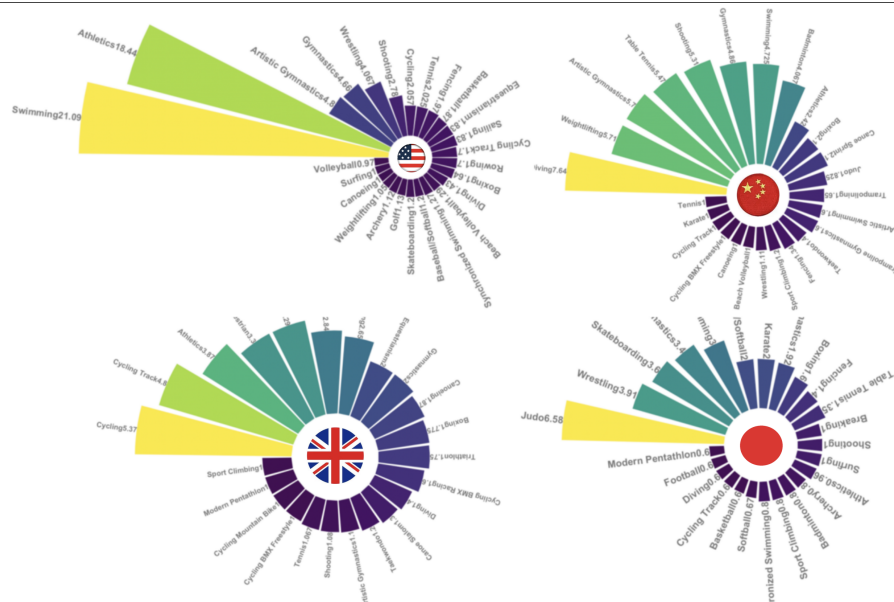


Figure 3: Rose diagram of each country's Discipline Advantage Level Index

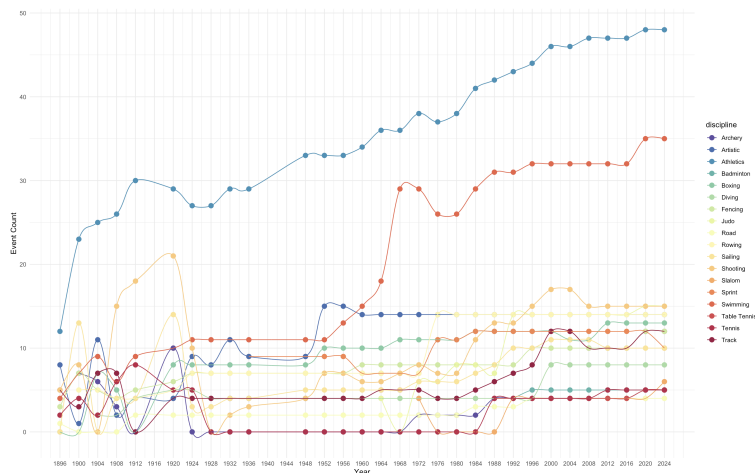


Figure 4: the Number of Events in Each Discipline's Line Chart

disciplines and the total number of medals won by the country. Therefore, for this country, there is motivation to increase the number of events in its advantageous disciplines to increase the chances of winning medals.

Next, we will explore the relationship between the significant changes in the number of these four disciplines and the total medal counts of their corresponding dominant countries. (Swimming and Athletics - US, Shooting and Diving - CHN).

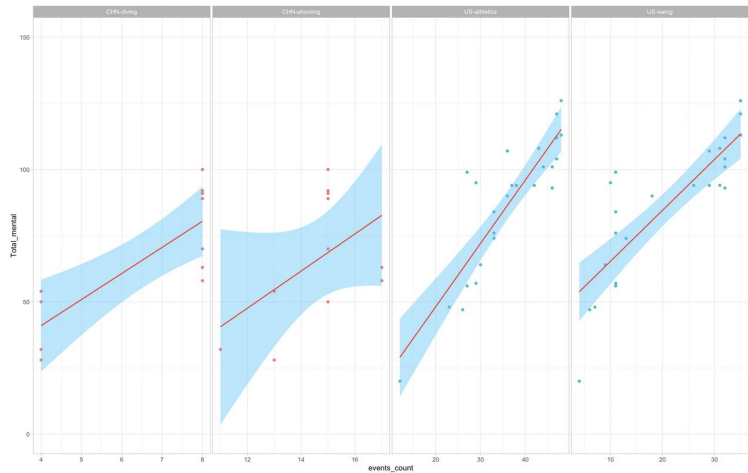
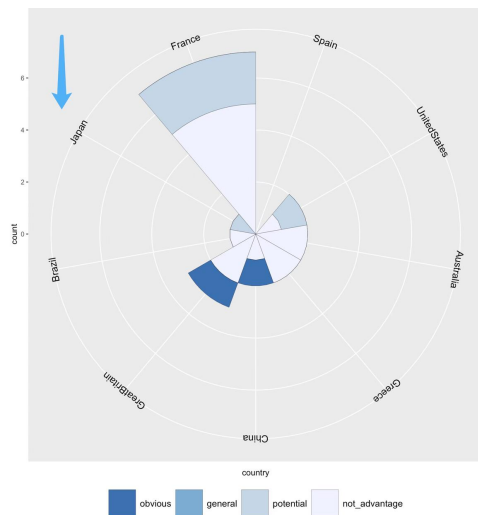


Figure 5: Scatter plot between the number of advantageous projects and the number of medals won by concepts



Rose diagram of the number of each host country's disciplines that decreased



Rose diagram of the number of each host country's disciplines that increased

We can clearly see that when the country acts as the host, the added events are mostly in their obvious advantageous disciplines, while the reduced ones are mostly in areas they are not good at. In addition to this, the host can also increase their medal advantage by adding projects with national characteristics. On the one hand, this behavior promotes the country's traditional culture, and on the other hand, it also increases the country's chances of winning championships. For example, Japan introduced a new project, karate, in the 2020 Olympics, and set up eight events under it. The introduction of this traditional Japanese project greatly increased the audience's enthusiasm and understanding of the project, and therefore also increased Japan's "gold medal points" on the medal table by 8, which could possibly win gold medals. That is to say, the host does indeed change the number of projects in their favor in terms of behavior, and then tries to achieve better medal-winning results.

5.2 Host effect

First, we need to investigate whether the total number of medals won by the country in the year it hosts the Olympics has increased. As shown in the figure above, almost all countries have seen an increase in the total number of medals won in the year they host the Olympics, especially before 1988, when this fluctuation is extremely obvious. Even after 1988, the host still has a certain advantage in the total number of medals won. Moreover, according to the line chart showing the total number of medals and the number of gold medals won by the eight countries that have hosted the Olympics in the past three decades, the host effect on the total number of medals will continue to exist, while for the number of gold medals, it is not the case, the host effect is only reflected in the year of hosting.

This provides us with insights into establishing a regression model. Considering all the above factors, including the host effect, the number of various medals (total medals, gold medals, silver medals, bronze medals) won in the previous Olympics, and the number of participants in the competitions, we establish a regression model with these five independent variables to predict the medal table for the next Olympics.

5.3 The construction process of the two-step random forest model

In the process of determining the model, we successively employed several classic modeling methods, including a linear regression, a Support Vector Machine (SVM) that accounts for non-linear transformations of features, a decision tree, and Random Forest with varying numbers of trees. Ultimately, we found that the Random Forest model with 10 decision trees outperformed all other models. Therefore, we decided to use the Random Forest model, as it typically offers higher predictive accuracy compared to other models. In terms of model robustness, the Random Forest effectively reduces overfitting and enhances stability and accuracy by aggregating the results of multiple decision trees.

Random Forest is an ensemble learning method that enhances model accuracy by constructing multiple decision trees and combining their predictions. Compared to a single decision tree, Random Forest typically handles data noise and overfitting more effectively, as it reduces the bias and variance of individual trees through the ensemble approach.

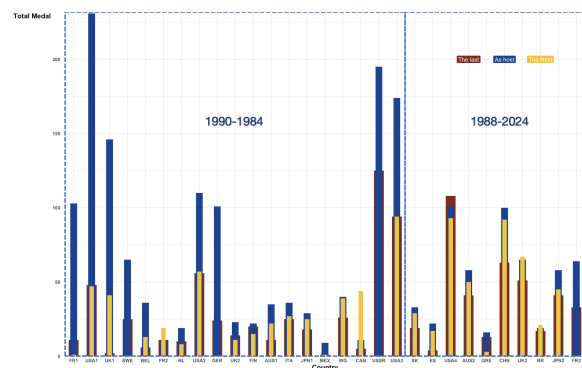


Figure 6: the Number of Total Medals Won by Union Soviet and Russia

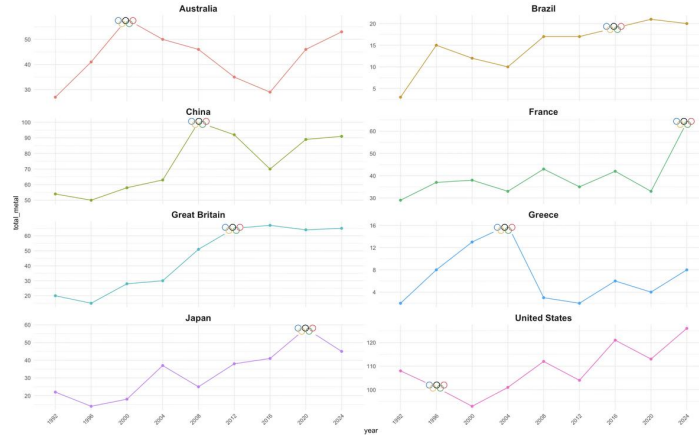


Figure 7: Flow Chart of Total Medals Won by Host Countries since 1992

In this study, we predict Olympic performance **using a two-stage Random Forest model**. We employ a two-step approach—first, we fit the number of medals obtained in the current year using known data. After determining the exact medal count for each team, we estimate the rational feasibility of each team winning their medals. When using the Random Forest model for regression, the selection and processing of independent variables are extremely important. We provide additional explanations below for the specific independent variables chosen in our regression model.

To have a clear introduction to the model, we need to have some further insight of the variables, especially the host effect and summary index.

Since whether a country is the host is not a continuous random variable, we define the host in the form of a categorical variable. The home-field advantage of the Olympic host country due to factors such as the country's cultivation of athletes and the encouragement from the home-crowd support that is, the Olympic Games held before the country hosts the Olympics. Moreover, according to the visual graphs above illustrate, this impact is likely to persist in the next Olympic cycle. In other words, the influence of a country hosting the Olympic Games can last for three Olympic Games, although the performance of the host country in the Olympic Games it hosts is more prominent in terms of data. This means that in our analysis of the host, we need to consider the host effect in the long run. In other words, the definition of the variable **host** (m represents the m th Olympic Games) should be as follows:

$$Host_{ij} = \begin{cases} 1, i \geq m \\ 0, i < m \end{cases} \quad (3)$$

Next, we need to discuss the variable **Summary_Index**. For a country, the total number of medals it wins is directly related to the number of medals it wins in each individual event. Therefore, we use the index of a country's advantageous projects in a certain field,

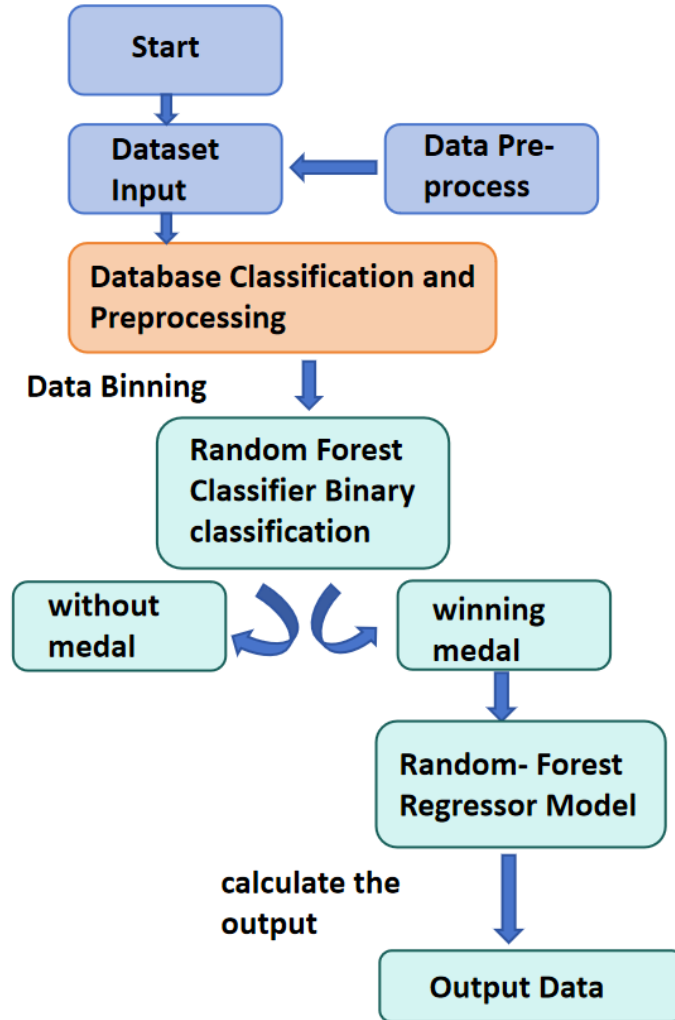


Figure 8: the Construction of Problem 1

which will be introduced shortly: Discipline Strength Level Index, to sum up the total content for each year respectively. And it could be calculate as follows :

$$SummaryIndex_j = \sum_k T_{kj} \quad (4)$$

Last , since the amount of medals definitely is related with the number of total medals ,so we use E_j as a variables in order to demonstrate its effect . Each decision tree is trained on a random subset of the original dataset, ensuring that each tree has some degree of diversity. The construction process of Random Forest includes the following steps:

Sampling: Conduct bootstrap sampling from the original dataset, which means randomly drawing samples to generate multiple training subsets. Each training subset will be used to train a decision tree.

	Gold _{Count}	Silver _{Count}	Bronze _{Count}
R ² Score	0.9531	0.9253	0.9477
RMSE	3.76	3.92	3.81

Table 8: Caption

5.4 the Prediction Result of 2028 Olympic Games

From the above model, we can calculate the following predicted results as shown in the figure below :

Compare the predicted values (the number of gold, silver, and bronze medals each participating country will win in 2028) with the known medal counts of each country in 2024. If all these values are greater than or equal to those of 2024, it is defined as performing better than the previous year. Conversely, if at least two of these values are worse than those of the previous year, it is defined as performing worse than the previous year.

Below are tow charts regarding answers we figure out for 2028. Before conducting the aforementioned random forest regression operation, we first used the random forest binary classification operation, that is, predicting whether each country can win medals in the next Olympic Games. In the random forest model we built, the binary classification

NOC	gold	80% upper confidence interval	80% lower confidence interval
United States	42	30.7	39.4
China	36	29.3	43.7
France	20	12.8	28
Australia	17	10.6	23.2
Germany	15	6.9	23.2
Japan	15	6.9	23.3
Great Britain	13	5.9	21
Italy	13	8.7	17.9

Table 9: the predicted number of golds in 2028 Los Angeles Olympic Games

NOC	silver	80% upper confidence interval	80% lower confidence interval
United States	53	17.7	45
China	23	17	29.1
France	22	16	28.9
Australia	16	11.7	20.5
Germany	14	7.1	20.2
Japan	13	7.2	19.2
Great Britain	15	6.4	22.9
Italy	12	7.3	17.4

Table 10: the predicted number of silvers in 2028 Los Angeles Olympic Games

NOC	bronze	80% upper confidence interval	80% lower confidence interval
United States	43	22.9	37.0
China	27	21.4	33
France	19	13.1	24.6
Australia	15	9.6	20.5
Germany	19	11.4	26.3
Japan	19	11.6	26.8
Great Britain	14	4.2	22.8
Italy	14	9.5	19.3

Table 11: the predicted number of bronzes in 2028 Los Angeles Olympic Games

NOC	total medals	80% upper confidence interval	80% lower confidence interval
United States	138	129.8	149.3
China	86	67.7	105.8
France	61	41.9	81.5
Australia	48	31.9	64.2
Germany	48	25.4	69.8
Japan	45	24.7	67
Great Britain	42	14.6	67.6
Italy	39	25.5	54.6

Table 12: the predicted number of total medals in 2028 Los Angeles Olympic Games

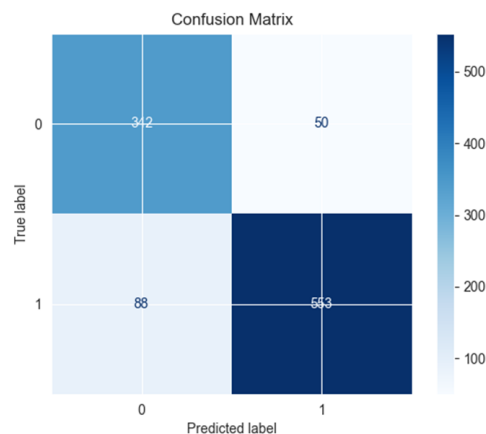
NOC	gold (2028)	gold (2024)	total medal (2028)	total medal (2024)
United States	43	40	138	126
Germany	15	12	48	33
Czechia	3	0	9	0
Nigeria	1	0	4	0

Table 13: the Countries that Performance Better

NOC	gold (2028)	gold (2024)	total medal (2028)	total medal (2024)
France	15	16	54	64
Australia	17	18	48	53
Great Britain	13	14	42	65
South Korea	9	13	23	32

Table 14: the Countries that Performance Worse

operation uses the number of gold, silver, and bronze medals won by each country in the previous Olympic Games, the number of participants in the projects, the summary_index, and whether medals were won in the previous Olympic Games as six labels for the independent variables. The first five labels are numerical variables, and whether medals were won in the previous Olympic Games is a categorical variable, using bool values for 0-1 transformation. The dependent variable is whether medals can be won in the next Olympic Games. According to the binary classification operation, we predict that in the next Olympic Games, a country that has never won medals in the Olympic Games will win medals for the first time. This country is Czechia, and the probability of them winning medals is 0.9265.



Based on the Confusion Matrix, the model's Precision is calculated to be 0.917, Recall is 0.863, and the F1 Score is 0.889. We can see that the classification effect of the binary classification operation is good and can be used to predict the medal-winning situations of various countries in the next Olympic Games.

Based on the ROC curves of the model on the training and test sets, we set the threshold of the random forest binary classifier to 0.9, and used this to predict the medal-winning situations of various countries in 2028. Finally, it was concluded that among the countries that have never won medals, the Czech Republic has a probability of 0.9265 of winning medals in the next Olympic Games, which exceeds the set threshold, and it can be judged that the country will win its first medal in the next edition.

6 Problem 2: Model 2

Two well-known factors that affect a country's performance on the Olympic medal table in the short term are the level of coaches and athletes. However, while the existence of nationality conversion for world-class athletes is quite hindered, coaches can actively choose the country they coach in different countries. This means that a pioneering "legendary" coach is very likely to greatly improve the performance of a team in a short time. In the history of the Olympics, there are no shortage of such examples. One of them is the legendary gymnastics coach Béla Károlyi, who has coached in Romania and the United States. Referring to historical literature on this coach, his coaching career can be described

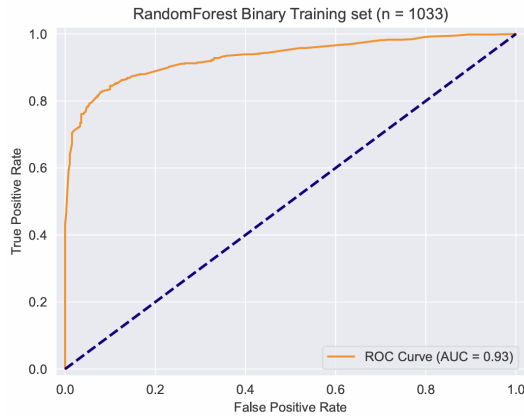


Figure 9: ROC of Trained Database

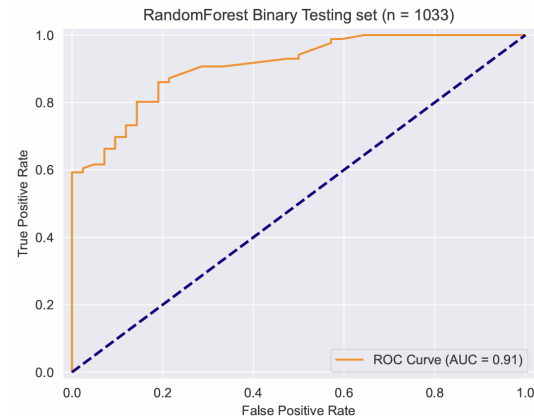


Figure 10: ROC of Test Database

as full of ups and downs. From 1974 to 1989, he first served as the national team coach, leading the once glorious Romanian gymnastics team. Ultimately, under his leadership, the Romanian team made significant progress at the Olympics. Subsequently, he went to the United States to establish his own gymnastics club, where he was able to fully exert his coaching talents, cultivating a large number of outstanding gymnasts for the United States, and creating the brilliant achievement of winning 10 gold medals, 10 silver medals, and 2 bronze medals at one Olympics.

6.1 Analysis of "Great Coach" Effect

Given the relative stability of the international environment over the past three decades and the fact that sporting strengths often have a legacy effect, we assume in our response to the first question that each country's strong sports disciplines remained constant from 1992 to 2024. Although this assumption is logical, there are exceptions. This is because many sports are skill-intensive. For countries with a weak foundation in a particular sport, hiring a highly professional coach can significantly improve their performance in that discipline. This phenomenon is referred to as the "great coach" effect.

Examine the data to identify the "great coach" phenomenon and assess its impact on the medal standings. Select three countries and determine the sports in which they should consider investing in a "great" coach, estimating the magnitude of this effect.

Regression Discontinuity Design (RDD), commonly referred to as "regression discontinuity," is one of the most popular quasi-experimental causal inference methods in empirical economic research. Thistlethwaite & Campbell (1960) first proposed regression discontinuity and used it to study the impact of scholarships on future academic achievement. After 1990, it began to be used by scholars in economic research, policy evaluation, and other fields. Hausman and Rapson first proposed Regression Discontinuity in Time (RDiT) in 2018[5].

RDD is a quasi-experimental design that does not involve random assignment of subjects. First, individuals receive a score, and if this score exceeds a known cutoff point, denoted as c , they are assigned to the treatment group to receive policy treatment; oth-

erwise, they are assigned to the control group. The RDiT method takes into account the time-series characteristics of the data, using the start time of the policy as the cutoff point, and sets the grouping variable based on the time interval to determine whether the explanatory variable jumps at the time point affected by the policy shock, further analyzing the causal relationship between the driving variable and the explained variable to assess the policy effect. This paper introduces this method to examine the impact of the "great coach" on medal counts.

Given the enormous impact that Béla Károlyi, who coached both Romania and the U.S. Women's teams, we analyze the effect of the "great coach" by considering the impact of this coach in both countries together.

Béla Károlyi served as the gymnastics coach for Romania from 1974 to 1980 and for the United States from 1981 to 2016. We select the years he began coaching as the breakpoints and define the following treatment variables:

$$D_j = \begin{cases} 1, j \geq 0 \\ 0, j < 0 \end{cases} \quad (5)$$

Among them, T_i is the forcing variable, representing the interval of Olympic Games between the time and the "critical point" when coaching began (for example, for the United States, 1981 is taken as the breakpoint, $T_i \leq 0$ indicates the time after 1981, and $T_i < 0$ indicates the time before 1981). D_i is the treatment variable, indicating whether the country is affected by the great coach effect. If $D_i=1$, it belongs to the experimental group, and if $D_i=0$, it belongs to the control group. The regression discontinuity model is designed as follows:

$$IE_{j,t} = \alpha + \delta D_{j,t} + \beta T_{j,t} + \varphi D_{j,t} T_{j,t} + \gamma X_{j,t} + \rho_j + \varepsilon \quad (6)$$

Among them, j represents the country, t represents different years, and $D_{j,t}$ is the treatment variable in this formula.

To examine the "great coach" effect, our group extracted the number of gymnastics gold medals won by the national teams of Romania and the United States before and after this legendary coach coached in these two countries, and used the regression discontinuity model to identify this sudden change.

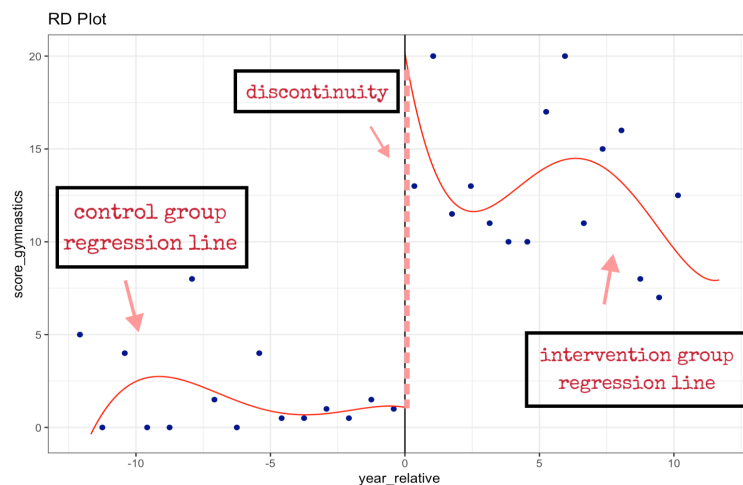
One of the prerequisites for using the regression discontinuity model is that the forcing variable must be exogenous. Most studies usually use McCrary's method to test whether the forcing variable has endogeneity. Since the forcing variable in the regression discontinuity in time framework is time, whether it is affected by the great coach effect depends on the year the coach started teaching, not a subjective decision. Therefore, the research sample near the breakpoint will not be manipulated by other factors, which can ensure the validity of the coefficient estimates.

It is also necessary to meet the premise that the covariates are continuously stable on both sides of the discontinuity. The test results for the continuity of the covariates are

as follows. The insignificant results indicate that the conditional density functions of all covariates do not have obvious jumps on both sides of the breakpoint, which means that the study of the great coach effect is valid.

Method	Coef.	Std. Err.	z	P> z	[95% C.I.]
Conventional	-20.684	12.931	-1.600	0.110	[-46.028 , 4.661]
Robust	-	-	-1.924	0.054	[-89.633 , 0.819]

It is necessary to test whether the number of gymnastics medals has changed significantly on both sides of the breakpoint. The figure below shows the third-order fitting results of the innovation efficiency of high-tech industries and the forcing variable.



The x-axis represents the interval of Olympic Games between the time and the "critical point" when coaching began. It can be seen that the number of gymnastics medals has a clear upward jump at the zero difference point, indicating that the level of gymnastics has increased to a higher degree after Béla Károlyi started coaching.

Next ,we need to do some result analysis ,

Method	Coef.	Std. Err.	z	P> z	[95% C.I.]
Conventional	13.504	1.830	7.380	0.000	[9.917 , 17.090]
Robust	-	-	4.160	0.000	[6.638 , 18.466]

1. The impact coefficient of an excellent coach on the total number of medals is 13.504 (i.e., $\gamma = 13.504$). This means that at the discontinuity, the presence of an excellent coach is associated with an increase of 13.504 in the total number of medals.
2. $z = 7.38$, used to test whether the coefficient is significantly different from zero. A higher z -value usually indicates stronger statistical significance.
3. $p = 0.000$, much smaller than the common significance level (such as 0.05), rejecting the null hypothesis, indicating that the impact of an excellent coach on the total number of medals is statistically significant.
4. The 95% confidence interval is $[9.917, 17.090]$, meaning that we have 95% confidence that the true impact of an excellent coach on the total number of medals falls within this range.
5. Robust: Under this method, the z -value is 4.160, and the P -value remains 0.000, indicating that even when considering heteroscedasticity or other robustness issues, the impact of an excellent coach remains significant.

Last but not the least ,we do sensitivity analysis through Robustness Check ,which is to say If the running variable is continuously distributed, it means that individuals do not have the ability to precisely manipulate the reference variable at the discontinuity, and the local randomization assumption holds, thereby ensuring that the samples on both sides of the discontinuity can represent the overall population at the discontinuity. Pseudo

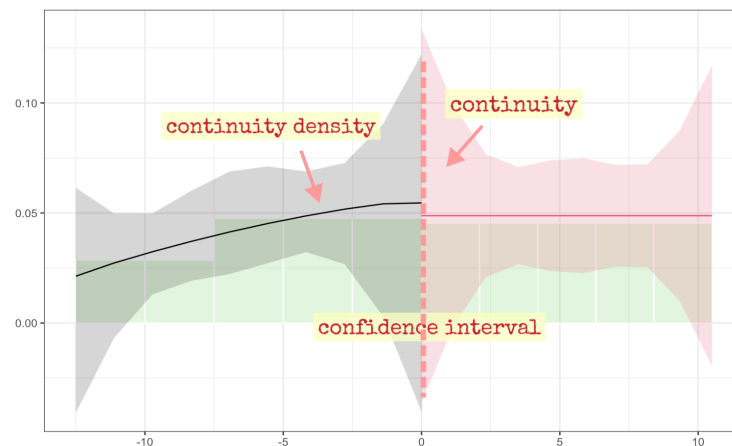


Figure 11: Running Variable Distribution

cutoff point test. At other positions of the running variable, such as 1, 2, -1, -2, estimate RDD using the same method, as shown in the figure below. The pseudo cutoff point intervention effect is zero (if it is found that the RDD estimator of the pseudo cutoff point is not zero, it means that our RDD design may be confounded by the influence of other unobserved factors, and the causal effect obtained may be caused by other unobserved confounding jumps, not entirely the effect of the intervention), which indicates that the



Figure 12: Pseudo cutoff point test

model is robust. From all these above ,the result of Robustness Check performances quite well , "great coach effect "is obvious,Béla Károlyi could efficiently increase the number of Olympic medal for this country , with an average of 18 more medals.

6.2 Take Advantage of the Effect of a "Great Coach"

Based on the above analysis, we can infer that the sports to be invested in are mainly those with a higher number of events set up annually and in which the country has potential advantages. We have selected Germany, Italy, and Australia for analysis.

- x_{j1} represents discipline advantage level index
- x_{j2} represents event_count (Given that many events have been withdrawn in recent years, the number of events is replaced by the average number of events in the 2020 and 2024 Olympics)

Perform k-means clustering algorithm analysis on the continuous variables x_1 and x_2 of the three countries .K-means clustering partitions the dataset into k clusters through iterative optimization, with each cluster represented by a centroid (Centroid). Its goal is to minimize the sum of squared errors (SSE) between data points within the cluster and the centroid, that is:

$$SSE = \sum_{i=1}^k \sum_{x \in c_i} \|x - \mu_i\|^2 \quad (7)$$

Where C_i represents the i_{th} cluster, μ_i represents the centroid of the i-th cluster, and x is the data point. After selecting 5 initial centroids, the process involves continuously iterating by assigning data points and updating centroids until the centroids no longer change. The results are shown in the figure below:

- Let the advantage level index of a certain country in a particular discipline be A.
- Let the highest advantage level index A_{max} in the global context, among different countries in that discipline.

And it can also be simplified as following ,

$$Result = 2(A_{max} - A) \quad (8)$$

And we have some suggestions as follows .

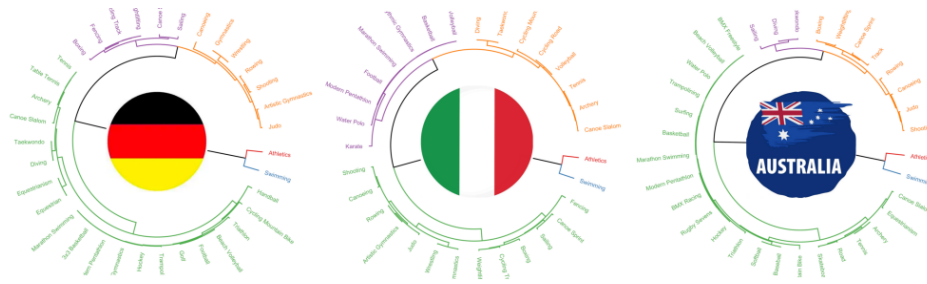


Figure 13: Enter Caption

1. Germany The orange section of disciplines meets the criteria of having a higher number of events set annually and the country has potential advantages. For example: Shooting, it is estimated that the number of medals increased after investment will be about 10.
2. Italy The green section of disciplines meets the criteria of having a higher number of events set annually and the country has potential advantages. For example: Rowing, it is estimated that the number of medals increased after investment will be about 7.
3. Australia The purple section of disciplines meets the criteria of having a higher number of events set annually and the country has potential advantages. For example: Diving, it is estimated that the number of medals increased after investment will be about 14.

7 More Insights

The International Olympic Committee has the following passage in the "Principles of Olympism" clause of the Olympic Charter: "Every person should have the possibility to practice sport without any form of discrimination and in the Olympic spirit, which requires mutual understanding, friendship, solidarity and fair competition." This is also known as the modern Olympic spirit.

It is the modern Olympic spirit that guides us to be faster, higher, and stronger, and to move towards a broader platform. However, based on the model training, we find that the host effect greatly affects a country's possibility of winning medals in the current year and even in the future. In the model, we observe that the host country has a tendency to increase its own obvious advantageous projects and reduce its own potential advantageous projects in order to achieve the effect of winning more medals. But this is not a good message for countries that cannot become the host. We understand the huge efforts that the host country needs to make to hold the Olympic Games, but there are still countries that cannot afford the expenses of hosting the Olympic Games, and may not even have a peaceful and stable environment. Should a country without a stable and moderately prosperous environment be at the back of the starting line?

We predict that the probability of the Refugee Olympic Team winning their first gold

medal in 2028 is 0.6156. Although this is greater than 0.5, due to the choice of threshold, we did not include them in the list of countries likely to win medals in the next Olympic Games. This makes me feel a bit regretful. Amidst the turmoil of war, they persist in sports and competition despite the pressures of survival, home crises, and other related issues. However, the arrangement of certain sports events may have pushed them further away from winning medals. **Therefore, I propose that the Olympic committees can consider giving non-host countries a greater say in the composition of sports events.**

At the same time, the model has statistically analyzed the number of Olympic Games participated in by different athletes and found that among a large number of athletes from developing and developed countries, the average sports career of athletes from developing countries is shorter than that of developed countries, which is regrettable. We also call on the Olympic committees to provide more support and help to athletes.

Currently, the EOR (Refugee Olympic Team) has competed in three Olympic Games. Their presence is an inspiration and encouragement not only for the athletes themselves but also for their hometowns and for people around the world who yearn for peace and development. We believe that the future Olympic arenas will continue to create more possibilities.

8 Conclusion

8.1 Strengths

- We introduced the Discipline Advantage Level Index variable into the model, which quantifies the sports advantages of various countries and better adapts the data to the mathematical model.
- The superposition of random forest binary classification and regression operations makes the model more continuous and interpretable, and also has higher computational efficiency, which can be extended to more complex problems with larger data volumes.
- In the process of studying the problem, we innovatively used Regression Discontinuity in Time (RDIT), which effectively dealt with the issue of time discontinuity. Moreover, after inspection, this regression discontinuity model has high applicability and robustness.

8.2 Weakness

- The use of the double random forest model requires higher data preprocessing, which may lead to a slightly longer dataset preparation time.
- The model is relatively idealized and needs to consider political, economic, and social factors to further promote the model.

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